

CSCI 5707 Fall 2023 – Homework 4 solutions

Remark 1. Please be consistent with the log notation. It is OK to use ‘log’ to denote base 10, natural log, and even log2. Except these three bases, please be explicit about the base.

Remark 2. Normalized relevance scores are always upper bounded by 1, as guaranteed by the Cauchy-Schwarz inequality in normed inner product space.

$$|\langle f, g \rangle| \leq \|f\| \|g\|, \|f\| = \langle f, f \rangle^{\frac{1}{2}}$$

Ex 27.2. Assume you are given a document database that contains SIX documents. Answer the following questions. (All answers use log base of 10.)

1. Show the result of creating an inverted file on the documents.

Term (Word)	Number of Documents	Documents
Auto	2	{1, 2}
Beetle	1	{6}
Car	2	{1, 6}
Computer	3	{2, 4, 5}
Honda	2	{1, 3}
IBM	2	{4, 5}
Manufacturer	2	{1, 4}
Navigation	2	{2, 3}
Personal	1	{5}
VW	1	{6}

2. Show the result of creating a signature file with a width of 5 bits. Construct your own hashing function that maps terms to bit positions

Hashing Function

Term (Word)	Hashed Value
Auto	10000
Beetle	10000
Car	01000
Computer	01000

Honda	00100
IBM	00100
Manufacturer	00010
Navigation	00010
Personal	00001
VW	00001

Document	Terms	Hash
1	car, manufacturer, Honda, auto	11110
2	auto, computer, navigation	11010
3	Honda, navigation	00110
4	Manufacturer, computer, IBM	01110
5	IBM, personal, computer	01101
6	car, Beetle, VW	11001

3. Evaluate the following boolean queries using the inverted file that you created: 'car', 'IBM' AND 'Computer', 'IBM' AND 'car', 'IBM' or 'auto', and 'IBM' AND 'computer' AND 'manufacturer'.

Using inverted file

- car -> returns {1,6}
- IBM AND computer -> returns {4,5}
 - 'IBM' appears in documents - Doc 4 and Doc 5
 - 'computer' appears in documents - Doc 2, Doc 4 and Doc 5
 - So their intersection - Doc 4 and Doc 5
- IBM AND car -> returns {}
 - 'IBM' appears in documents - Doc 4 and Doc 5
 - 'car' appears in documents - Doc 1, and Doc 6
 - So their intersection - None
- IBM OR auto -> returns {1, 2, 4, 5}
 - 'IBM' appears in documents - Doc 4 and Doc 5
 - 'auto' appears in documents - Doc 1, and Doc 2
 - So their union - Doc 1, Doc 2, Doc 4 and Doc 5
- IBM AND computer AND manufacturer -> returns {4}
 - 'IBM' appears in documents - Doc 4 and Doc 5
 - 'computer' appears in documents - Doc 2, Doc 4 and Doc 5

'manufacturer' appears in documents - Doc 1, and Doc 4

So their intersection - Doc 4

5. Consider the following ranked queries: 'car', 'IBM computer', 'IBM car', 'IBM auto', 'IBM computer manufacturer'

(a) Calculate the IDF for every term in the database.

Term (Word)	IDF
Auto	$\log(6/2) = \log(3) = 0.477$
Beetle	$\log(6/1) = \log(6) = 0.778$
Car	$\log(6/2) = \log(3) = 0.477$
Computer	$\log(6/3) = \log(2) = 0.301$
Honda	$\log(6/2) = \log(3) = 0.477$
IBM	$\log(6/2) = \log(3) = 0.477$
Manufacturer	$\log(6/2) = \log(3) = 0.477$
Navigation	$\log(6/2) = \log(3) = 0.477$
Personal	$\log(6/1) = \log(6) = 0.778$
VW	$\log(6/1) = \log(6) = 0.778$

(b) For each document, show its document vector.

	Auto	beetle	car	computer	honda	IBM	manufacturer	navigation	personal	VW
Doc 1	1	0	1	0	1	0	1	0	0	0
Doc 2	1	0	0	1	0	0	0	1	0	0
Doc 3	0	0	0	0	1	0	0	1	0	0
Doc 4	0	0	0	1	0	1	1	0	0	0
Doc 5	0	0	0	1	0	1	0	0	1	0
Doc 6	0	1	1	0	0	0	0	0	0	1

Document Vectors (TF-IDF) are as follows:

	Auto	beetle	car	computer	honda	IBM	manufacturer	navigation	personal	VW
Doc 1	0.477	0	0.477	0	0.477	0	0.477	0	0	0
Doc 2	0.477	0	0	0.301	0	0	0	0.477	0	0
Doc 3	0	0	0	0	0.477	0	0	0.477	0	0
Doc 4	0	0	0	0.301	0	0.477	0.477	0	0	0
Doc 5	0	0	0	0.301	0	0.477	0	0	0.778	0
Doc 6	0	0.778	0.477	0	0	0	0	0	0	0.778

(c)

Query Vectors (TF-IDF) are as follows:

	auto	beetle	Car	computer	honda	IBM	manufacturer	navigation	personal	VW
Query 1	0	0	0.477	0	0	0	0	0	0	0
Query 2	0	0	0	0.301	0	0.477	0	0	0	0
Query 3	0	0	0.477	0	0	0.477	0	0	0	0
Query 4	0.477	0	0	0	0	0.477	0	0	0	0
Query 5	0	0	0	0.301	0	0.477	0.477	0	0	0

Doc length (Used for length normalization)

	Length = $\sqrt{\sum_{k=1}^t w_k^2}$
Doc 1	0.954
Doc 2	0.739
Doc 3	0.675
Doc 4	0.739
Doc 5	0.961
Doc 6	1.199

Query length

	Length
Q 1	0.477
Q 2	0.564
Q 3	0.675
Q 4	0.675
Q 5	0.739

$$\text{Relevance}(Q,D) = \sum_{j=1}^t w_{qj} * w_{dj}$$

$$\text{Normalized relevance} = \frac{\text{relevance}(Q,D)}{\text{length}(Q)*\text{length}(D)}$$

Relevance without length normalization:

Sim(Q,D)	Q1	Q2	Q3	Q4	Q5
Doc 1	0.228	0	0.228	0.228	0.228
Doc 2	0	0.091	0	0.228	0.091
Doc 3	0	0	0	0	0
Doc 4	0	0.318	0.228	0.228	0.546
Doc 5	0	0.318	0.228	0.228	0.318
Doc 6	0.228	0	0.228	0	0

Relevance with length normalization:

Sim(Q,D)	Q1	Q2	Q3	Q4	Q5
Doc 1	0.5	0	0.354	0.354	0.323

Doc 2	0	0.217	0	0.457	0.166
Doc 3	0	0	0	0	0
Doc 4	0	0.764	0.457	0.457	1
Doc 5	0	0.587	0.351	0.351	0.448
Doc 6	0.398	0	0.281	0	0

(f)

For each term, tf will be 10 times larger while idf remains unchanged. Therefore, tf-idf vector for document i , $d'_i = 10d_i$. The query vectors q_j stay the same. Let $\langle \cdot, \cdot \rangle$ denotes the inner product of the Euclidean space.

Unnormalized score:

$$s'(i, j) = \langle d'_i, q'_j \rangle = \langle 10d_i, q_j \rangle = 10 \langle d_i, q_j \rangle = 10s(i, j)$$

Normalized score:

$$s'_{normalized}(i, j) = \frac{10 \langle d_i, q_j \rangle}{10^{\frac{1}{2}} \langle d_i, d_i \rangle^{\frac{1}{2}} \cdot 10^{\frac{1}{2}} \langle q_j, q_j \rangle^{\frac{1}{2}}} = \frac{\langle d_i, q_j \rangle}{\langle d_i, d_i \rangle^{\frac{1}{2}} \cdot \langle q_j, q_j \rangle^{\frac{1}{2}}} = s_{normalized}(i, j)$$

Ex 27.4

1. If you were given a budget of \$500,000 dollars for purchasing machines, and were required to index all 10 billion pages, could you do it?

\$500,000 is equivalent to $500000/1000 = 500$ machines. This is equivalent to $500 * 10$ Million pages = $5 * 10^9$ pages. Which is equivalent to 5 billion pages. However the requirement is for 10 billion pages and hence we cannot do it.

2. What is the minimum budget to index all pages? If you assume that each query can be answered by looking at data in just one (10 million page) partition, and that queries are uniformly distributed across partitions, what peak load (in number of queries per second) can such a cluster handle?

Minimum cost would be equivalent to \$1,000,000 = 1 Million Dollars. This would allow us to have enough machines to index around 10 billion pages in 10 million page partitions on 1000 machines. The server would be able to handle around $200 * 1000 = 200,000$ queries per second.

3. How would your answer to the previous question change if each query, on average, accessed two partitions?

Then the answer would be around **100,000 queries per second**. Because two machines together are expected to respond to 200 queried per second.

4. **What is the running budget required to handle the desired load of 500 million hits per day if all queries are on a single partition? Assume that queries are uniformly distributed with respect to time of day.**

Assumption - That the facility already has stored the 10 billion pages.

500 million hits per day = 5788 queries per second. Considering they are using one partition each, we need $5788/200$ machines = 29 machines.

So the budget = $29 \times 1000 = \text{\$29,000}$

If we do want to account for the 10 billion pages, the budget should be \$1 million, which will give us more machines that needed to compute the queries while making sure that the machines are able to save all the 10 billion pages of indexed data.

5. **Would your answer to the previous question change if the number of queries per day went up to 5 billion hits per day? How would it change if the number of pages went up to 100 billion?**

(a) Yes, the answer will change as follows:

5 billion hits per day = 57880 queries per second. Considering they are using one partition each, we need $57880/200$ machines = 290 machines.

So the budget = $290 \times 1000 = \text{\$290,000}$

(b) Minimum budget for 100 billion pages:

Pages handled by 1 machine = 10 million

Machines required for 100 billion pages = $(100 \times 10^9) / (10 \times 10^6) = 10,000$

So the budget = $10,000 \times 1000 = \text{\$10 million}$

6. **Assume that each query accesses just one partition, that queries are uniformly distributed across partitions, but that at any given time, the peak load on a partition is upto 10 times the average load. What is the minimum budget for purchasing machines in this scenario?**

Assumption : That we have already been able to index all the pages (10 Billion)

- Assuming that the number of hits per day = 500m,
hits per day / $(60 \times 60 \times 24) = 5788$ hits per second in the worst scenario
- The peak would be 57880 hits in a second (10x).
- So, $57880 / 200 = 290$ machines needed
- The budget would be $290 \times 1000 = \text{\$290,000}$

7. **Take the cost for machines [take the previous question and multiply it by 10 to reflect the costs of maintenance, administration, network bandwidth, etc. This amount is your annual cost of operation. Assume that you charge advertisers 2 cents per page. What fraction of your inventory (i.e., the total number of pages that you serve over the course of a year) do you have to sell in order to make a profit?**

The cost of machines after accounting for the other costs → \$290,000

(It is OK to assume a different cost.)

- Number of ads pages = $\text{\$290,000} / 0.02 = 14,500,000 = 1.45 \times 10^8$ in one year

- We have 10^{10} pages. So the fraction needed to make a profit
 $= (1.45 * 10^8)/10^{10} = 1.45/100 = 1.45\%$

Ex 28.1 (most answers can be found in the book. Also fine to argue from a math perspective.)

1. spatial data assume some geometric structures, while it is unknown if nonspatial data can be embedded into metric spaces.
2. points have no volume. Assume natural topology in Euclidean space, the neighborhood of one point, no matter how small the radius is, never contains points other than itself. Pick arbitrary points in a regional object, this does not hold.
3. omitted.
4. Any reasonable answers.
5. Any reasonable answers. E.g. nearest neighbor queries in high dimensional space.
6. omitted.
7. Both are space-filling curves. Hilbert curve is preferred when you want to retain the local topological structure of the original data as much as possible. Namely, a neighborhood of points will still stay together/close enough, under Hilbert curve ordering. However, the jump in Z-order will break the structure.

Ex 28.5 (Consider what a single query can do.)

Any reasonable queries. Answers not unique.

- (1)-(4) omitted.
- (5) Query regions that contain the intersection of R1 and R2.
- (6) Query regions that contain R3 and R5.
- (7) Since R3 and R4 do not overlap with R5, any region search that contains both of them will do.
- (8) Impossible. Since $R3 \cap R4 \neq \emptyset$